

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
12.	(a)			carbonate (1) react with acid to release carbon dioxide (1)	1	1		2		2
				Indicative content A lead nitrate or lead ethanoate B potassium iodide C copper(II) oxide D magnesium carbonate E barium carbonate • mixing A and B gives yellow precipitate so one must contain lead ions and the other iodide ions • C gives blue solution so contains copper(II) ions • C is insoluble but doesn't fizz with acid so oxide or hydroxide • B gives a white solid and brown solution with C – suggesting B contains iodide so A must contain lead ions • only soluble lead salt is lead nitrate (or lead ethanoate)				6	6	6

			5-6 marks Identifies six ions correctly (in addition to the carbonates); bright yellow precipitate is used to identify the presence of both iodide ions and lead(II) ions; the presence of copper(II) ions is linked to either the pale blue colour or the reaction with iodide. <i>The candidate constructs a relevant, coherent and logically structured account including key elements of the indicative content. A sustained and substantiated line of reasoning is evident and scientific conventions and vocabulary is used accurately throughout.</i> 3-4 marks Identifies four ions correctly (in addition to the carbonates); valid reason is given for at least two of copper(II) ions, iodide ions and lead(II) ions <i>The candidate constructs a coherent account including many of the key elements of the indicative content. Some reasoning is evident in the linking of key points and use of scientific conventions and vocabulary is generally sound.</i> 1-2 marks Identifies three ions correctly (in addition to the carbonates); valid reason is given for at least one of copper(II) ions, iodide ions and lead(II) ions <i>The candidate attempts to link relevant points from the indicative content. Coherence is limited by omission and/or inclusion of irrelevant material. There is some evidence of appropriate use of scientific conventions and vocabulary.</i> 0 marks <i>The candidate does not make any attempt or give an answer worthy of credit.</i>
			Question 12 total 116808